

Dhruv Soni

sonidhruv1272@gmail.com | Personal Website | github.com/sonidhruv1272 | 647-622-6248

EDUCATION

York University

Bachelor Of Engineering In Computer Engineering (Co-op)

Sep 2024 - April 2028 (Expected)

Toronto, Canada

EXPERIENCE

Embedded Systems Developer, YorkU Robotics Society (YURS)

Sep 2024 - Present

York University

Toronto, Canada

- Developed autonomous Sumobot system in **C++ (Arduino IDE)**, improving competition win-rate by **35%** through optimized control logic and tuning.
- Implemented **ultrasonic + infrared sensor fusion**, reducing collision/edge detection errors by **40%** across test environments.
- Optimized **motor control loops** and timing precision, reducing response latency from **120ms to 70ms** during real-time matches.

Hardware & Embedded Systems Developer, IEEE YorkU Student Branch

Sep 2025 - Present

York University

Remote

- Designed and debugged **PCB prototypes** for student engineering builds, improving circuit stability and reducing failure rate by **25%**.
- Programmed microcontroller-based systems using **Arduino C/C++** for sensor integration and embedded control applications.
- Supported **10+ lab builds** involving circuit validation, soldering, debugging, and system-level hardware testing.

PROJECTS

Fingerprint Door Lock System

Nov 2024

- Built biometric authentication system using Arduino and fingerprint sensor with **98% verification accuracy** in testing, developing **embedded C++** logic for fingerprint matching, enrollment, and servo-based locking control.
- Programmed servo-based locking mechanism in **embedded C++**, reducing unlock/response time (**<1s latency**) via optimized PWM control, timing synchronization, and embedded system tuning.
- Implemented input validation and authentication logic to minimize false positives and unauthorized access attempts, improving system security and reducing invalid authentication events by **25%**.

Smart Doorbell System (ESP32-CA + Blynk IoT)

Jan 2025

- Developed IoT smart doorbell using **ESP32-CAM** with real-time **Wi-Fi notifications (<2s latency)**, implementing **embedded firmware** for event detection, cloud sync, and low-latency mobile communication.
- Built embedded firmware in **C++** for event-driven triggering and cloud communication via Blynk platform, enabling real-time IoT notifications with reliable Wi-Fi data transmission and **<2s response latency**.
- Integrated remote monitoring dashboard enabling live alerts and device control across mobile and web interfaces, improving responsiveness and reducing device management latency by **40%** through real-time synchronization.

PCB Design

Sep 2025

- Designed multi-layer PCB layouts using **CAD tools**, improving routing efficiency and reducing **signal noise by 30%** through optimized component placement, trace routing, and power distribution techniques.
- Simulated circuit behavior pre-fabrication, identifying and correcting **10+ design issues** before implementation by validating signal integrity, power distribution, and component connectivity.
- Applied power and component placement strategies to improve circuit stability and thermal performance by **30%**.

TECHNICAL SKILLS

Languages: Python, Matlab, Java, JavaScript, C/C++

Embedded/Hardware: Arduino, ESP32, Microcontrollers, Circuit Design, PCB Design, Smart Doorbell & IoT System Integration, Ultrasonic & IR Sensors, Servo & Motor Control, Hardware Debugging

Design/Developer Tools: Python Software Foundation, PyCharm, PCB CAD Software, Arduino IDE, ESP-IDF, MATLAB, IntelliJ, Eclipse, GitHub

Libraries/Frameworks: NumPy, pandas, Matplotlib, Flask, FastAPI, JUnit

Systems/Troubleshooting: BIOS Configuration, OS Installation, Driver Setup, Hardware-Software Integration, Component-Level Troubleshooting